

Gene sets in R

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Dependencies

This document depends on the following packages:

```
library(devtools)
library(Biobase)
library(goseq)
library(DESeq2)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools", "MatrixEQTL"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase", "goseq", "DESeq2"))
```

Download the data

Here we are going to follow along with the tutorial on goseq (<http://www.bioconductor.org/packages/release/bioc/vignettes/goseq/inst/doc/goseq.pdf>). You can see what genomes are supported by this package

```
head(supportedGenomes())
```

```
##          db species      date              name
## 1    hg38   Human Dec. 2013  Genome Reference Consortium GRCh38
## 2    hg19   Human Feb. 2009  Genome Reference Consortium GRCh37
## 3    hg18   Human Mar. 2006              NCBI Build 36.1
## 4    hg17   Human May 2004              NCBI Build 35
## 5    hg16   Human Jul. 2003              NCBI Build 34
## 6 vicPac2  Alpaca Mar. 2013  Broad Institute Vicugna_pacos-2.0.1
##
AvailableGeneIDs
## 1
## 2                                ccdsGene,ensGene,exoniphy, geneSymbol,k
nownGene,nscanGene,refGene,xenoRefGene
## 3  acembly,acescan,ccdsGene,ensGene,exoniphy, geneSymbol,geneid,genscan,knownGene,knownGeneOld3,refGene,sgpGene,sibGene,xenoRefGene
## 4 acembly,acescan,ccdsGene,ensGene,exoniphy, geneSymbol,geneid,genscan,knownGene,refGene,sgpGene,vegaGene,vegaPseudoGene,xenoRefGene
## 5                                acembly,ensGene,exoniphy, geneSymbol,geneid,genscan,knownGene,refGene,sgpGene
## 6
```

```
head(supportedGeneIDs())
```

```
##          db          track          subtrack          GeneID
## 1    knownGene      UCSC Genes              <NA>      Entrez Gene ID
## 2 knownGeneOld3 Old UCSC Genes              <NA>
## 3      ccdsGene          CCDS              <NA>
## 4      refGene      RefSeq Genes              <NA>      Entrez Gene ID
## 5 xenoRefGene      Other RefSeq              <NA>
## 6      vegaGene      Vega Genes Vega Protein Genes HAVANA Pseudogene ID
##
AvailableGenomes
## 1
hg16,hg17,hg18,hg19,mm7,mm8,mm9,rn3,rn4
## 2
hg18
## 3
hg17,hg18,hg19,mm8,mm9
## 4
bosTau2,bosTau3,bosTau4,canFam1,canFam2,ce2,ce4,ce6,ci1,ci2,danRer3,danRer4,danRer5,danRer6,dm1,dm2,dm3,
equCab1,equCab2,felCat3,gallGal2,gallGal3,hg16,hg17,hg18,hg19,mm7,mm8,mm9,monDom4,monDom5,ornAna1,oryLat2,panTro2,ponAbe2,rheMac2,rn3,rn4,
strPur1,strPur2,taeGut1,xenTro2
## 5 anoCar1,aplCal1,braFlo1,caeJap1,caePb1,caePb2,caeRem2,caeRem3,calJac1,canFam1,canFam2,cavPor3,cb1,cb3,ce4,ce6,ci1,ci2,danRer6,dp2,dp3,droAna1,droAna2,droEre1,droGri1,droMoj1,droMoj2,droPer1,droSec1,droSim1,droVir1,droVir2,droYak1,droYak2,equCab2,felCat3,gallGal3,hg17,hg18,hg19,loxAfr3,mm7,mm8,mm9,monDom4,monDom5,ornAna1,oryLat2,panTro1,panTro2,petMar1,ponAbe2,priPac1,rheMac2,rn3,rn4,
strPur1,strPur2,taeGut1
## 6
danRer5,hg17
```

An example of a goseq analysis

Load the data

Here we load the example from the `goseq` package.

```
temp_data = read.table(system.file("extdata", "Li_sum.txt",  
                                package="goseq"), sep="\t",  
                      header=TRUE,  
                      stringsAsFactors=FALSE)  
  
expr= temp_data[,-1]  
rownames(expr) = temp_data[,1]  
expr = expr[rowMeans(expr) > 5,]  
grp=factor(rep(c("Control", "Treated"), times=c(4,3)))  
pdata = data.frame(grp)
```

Perform a differential expression analysis

Now we perform a differential expression analysis for the group variable with `DESeq2`

```
de = DESeqDataSetFromMatrix(expr, pdata, ~grp)  
de_fit = DESeq(de)  
de_results = results(de_fit)
```

Get the differentially expressed genes after FDR correction

```
genes = as.integer(de_results$padj < 0.05)  
not_na = !is.na(genes)  
names(genes) = rownames(expr)  
genes = genes[not_na]
```

Pick the right genome

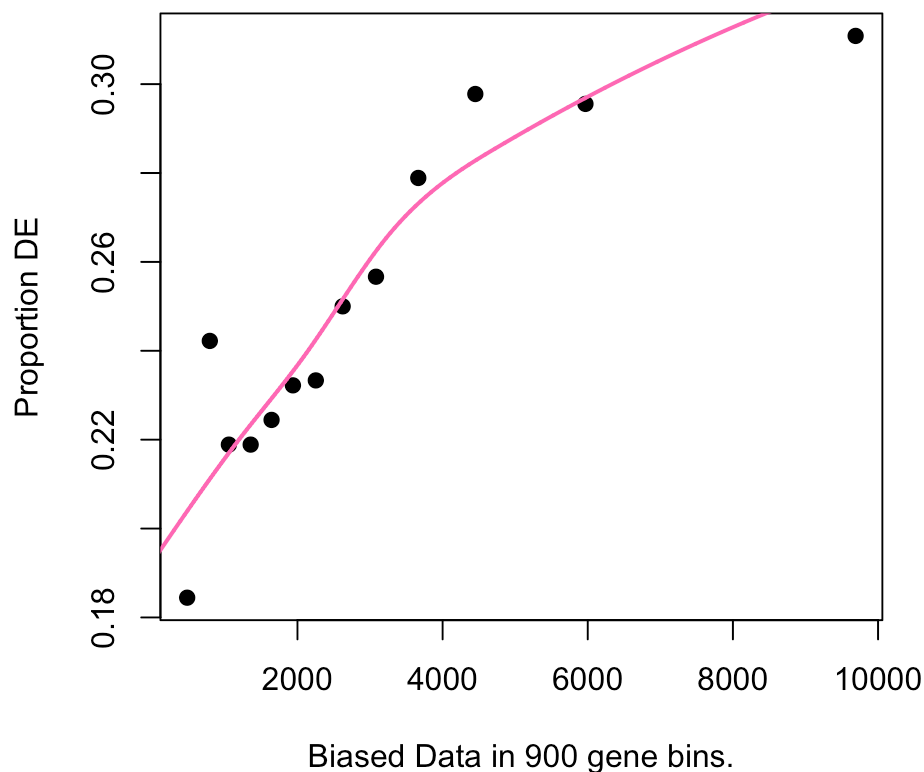
Here we look at some of the automatically supported genomes

```
head(supportedGenomes(), n=12)[,1:4]
```

##	db	species	date	name
## 1	hg38	Human	Dec. 2013	Genome Reference Consortium GRCh38
## 2	hg19	Human	Feb. 2009	Genome Reference Consortium GRCh37
## 3	hg18	Human	Mar. 2006	NCBI Build 36.1
## 4	hg17	Human	May 2004	NCBI Build 35
## 5	hg16	Human	Jul. 2003	NCBI Build 34
## 6	vicPac2	Alpaca	Mar. 2013	Broad Institute Vicugna_pacos-2.0.1
## 7	vicPac1	Alpaca	Jul. 2008	Broad Institute VicPac1.0
## 8	dasNov3	Armadillo	Dec. 2011	Broad Institute DasNov3
## 9	otoGar3	Bushbaby	Mar. 2011	Broad Institute OtoGar3
## 10	papHam1	Baboon	Nov. 2008	Baylor College of Medicine HGSC Pham_1.0
## 11	papAnu2	Baboon	Mar. 2012	Baylor College of Medicine Panu_2.0
## 12	felCat5	Cat	Sep. 2011	ICGSC Felis_catus-6.2

We need to set up a weighting function for all the genes in that genome

```
pwf=nullp(genes,"hg19","ensGene")
```



```
head(pwf)
```

```
##           DEgenes bias.data      pwf
## ENSG00000124208      0      1978 0.2362794
## ENSG00000224628      0      1510 0.2265504
## ENSG00000125835      0       954 0.2147844
## ENSG00000125834      0      6013 0.2972752
## ENSG00000197818      0      6295 0.2996675
## ENSG00000101181      0      2644 0.2520835
```

Perform the enrichment analysis parametrically

Here we use a parametric test to look for differences in enrichment with respect to different categories. This is the 2 x 2 table approach. You can also use random sampling to define the null distribution by setting the parameters `method="Sampling"` and `repnt=1000` for the number of null repetitions.

```
G0.wall=goseq(pwf,"hg19","ensGene")
head(G0.wall)
```

```
##           category over_represented_pvalue under_represented_pvalue
## 9656 G0:0044763      2.243186e-18      1
## 9578 G0:0044421      2.244623e-18      1
## 2058 G0:0005576      3.691127e-18      1
## 6756 G0:0031982      6.248427e-18      1
## 6761 G0:0031988      1.472125e-17      1
## 9636 G0:0044699      5.082258e-17      1
##           numDEInCat numInCat      term ontology
## 9656      1750      5932 single-organism cellular process BP
## 9578      616      1798      extracellular region part CC
## 2058      654      1930      extracellular region CC
## 6756      669      1984      vesicle CC
## 6761      651      1929      membrane-bounded vesicle CC
## 9636      1894      6537      single-organism process BP
```

Limiting yourself to a single category you are interested in

Suppose there is a particular category or function you are interested in. You can limit to just that category

```
G0.MF=goseq(pwf,"hg19","ensGene",test.cats=c("G0:MF"))
head(G0.MF)
```

```
##          category over_represented_pvalue under_represented_pvalue
## 909  GO:0005198          5.318741e-08          1.0000000
## 1004 GO:0005515          2.398716e-05          0.9999808
## 2566 GO:0044822          2.639927e-05          0.9999810
## 205  GO:0003723          6.656537e-05          0.9999495
## 1941 GO:0023023          1.045388e-04          0.9999963
## 1942 GO:0023026          1.045388e-04          0.9999963
##          numDEInCat numInCat          term ontology
## 909          129      334      structural molecule activity      MF
## 1004          1677     6006          protein binding      MF
## 2566          322     1024      poly(A) RNA binding      MF
## 205          395     1297          RNA binding      MF
## 1941           8        9      MHC protein complex binding      MF
## 1942           8        9 MHC class II protein complex binding      MF
```

More information

Gene set and other enrichment analyses are widely used to make sense of genomic results. Here are a couple of good places to start

- geneontology website (<http://geneontology.org/>)
- goseq vignette (<http://www.bioconductor.org/packages/release/bioc/vignettes/goseq/inst/doc/goseq.pdf>)
- seqGSEA package (<http://www.bioconductor.org/packages/release/bioc/html/SeqGSEA.html>)
- An early gene set enrichment paper (<http://www.pnas.org/content/102/43/15545.full>)

Session information

Here is the session information

```
devtools::session_info()
```

```

## setting value
## version R version 3.2.1 (2015-06-18)
## system x86_64, darwin10.8.0
## ui RStudio (0.99.447)
## language (EN)
## collate en_US.UTF-8
## tz America/New_York
##
## package * version date
## acepack 1.3-3.3 2014-11-24
## annotate 1.46.1 2015-07-11
## AnnotationDbi * 1.30.1 2015-04-26
## assertthat 0.1 2013-12-06
## BiasedUrn * 1.06.1 2013-12-29
## Biobase * 2.28.0 2015-04-17
## BiocGenerics * 0.14.0 2015-04-17
## BiocInstaller * 1.18.4 2015-07-22
## BiocParallel 1.2.20 2015-08-07
## biomaRt 2.24.0 2015-04-17
## Biostrings 2.36.3 2015-08-12
## bitops 1.0-6 2013-08-17
## bladderbatch * 1.6.0 2015-08-26
## broom * 0.3.7 2015-05-06
## caTools 1.17.1 2014-09-10
## cluster 2.0.3 2015-07-21
## colorspace 1.2-6 2015-03-11
## corpcor 1.6.8 2015-07-08
## curl 0.9.2 2015-08-08
## DBI * 0.3.1 2014-09-24
## dendextend * 1.1.0 2015-07-31
## DESeq2 * 1.8.1 2015-05-02
## devtools * 1.8.0 2015-05-09
## digest 0.6.8 2014-12-31
## dplyr * 0.4.3 2015-09-01
## edge * 2.1.0 2015-09-06
## evaluate 0.7.2 2015-08-13
## foreign 0.8-66 2015-08-19
## formatR 1.2 2015-04-21
## Formula * 1.2-1 2015-04-07
## futile.logger 1.4.1 2015-04-20
## futile.options 1.0.0 2010-04-06
## gdata 2.17.0 2015-07-04
## genefilter * 1.50.0 2015-04-17
## geneLenDataBase * 1.4.0 2015-09-06
## geneplotter 1.46.0 2015-04-17
## GenomeInfoDb * 1.4.2 2015-08-15
## GenomicAlignments 1.4.1 2015-04-24
## GenomicFeatures 1.20.2 2015-08-14
## GenomicRanges * 1.20.5 2015-06-09
## genstats * 0.1.02 2015-09-05
## ggplot2 * 1.0.1 2015-03-17
## git2r 0.11.0 2015-08-12

```

##	GO.db	3.1.2	2015-09-06
##	goseq	* 1.20.0	2015-04-17
##	gplots	* 2.17.0	2015-05-02
##	gridExtra	2.0.0	2015-07-14
##	gtable	0.1.2	2012-12-05
##	gtools	3.5.0	2015-05-29
##	highr	0.5	2015-04-21
##	HistData	* 0.7-5	2014-04-26
##	Hmisc	* 3.16-0	2015-04-30
##	htmltools	0.2.6	2014-09-08
##	httr	1.0.0	2015-06-25
##	IRanges	* 2.2.7	2015-08-09
##	KernSmooth	2.23-15	2015-06-29
##	knitr	* 1.11	2015-08-14
##	lambda.r	1.1.7	2015-03-20
##	lattice	* 0.20-33	2015-07-14
##	latticeExtra	0.6-26	2013-08-15
##	lazyeval	0.1.10	2015-01-02
##	limma	* 3.24.15	2015-08-06
##	lme4	1.1-9	2015-08-20
##	locfit	1.5-9.1	2013-04-20
##	magrittr	1.5	2014-11-22
##	MASS	* 7.3-43	2015-07-16
##	Matrix	* 1.2-2	2015-07-08
##	MatrixEQTL	* 2.1.1	2015-02-03
##	memoise	0.2.1	2014-04-22
##	mgcv	* 1.8-7	2015-07-23
##	minqa	1.2.4	2014-10-09
##	mnormt	1.5-3	2015-05-25
##	munsell	0.4.2	2013-07-11
##	nlme	* 3.1-122	2015-08-19
##	nloptr	1.0.4	2014-08-04
##	nnet	7.3-10	2015-06-29
##	org.Hs.eg.db	* 3.1.2	2015-07-17
##	plyr	1.8.3	2015-06-12
##	preprocessCore	* 1.30.0	2015-04-17
##	proto	0.3-10	2012-12-22
##	psych	1.5.6	2015-07-08
##	qvalue	* 2.0.0	2015-04-17
##	R6	2.1.1	2015-08-19
##	RColorBrewer	1.1-2	2014-12-07
##	Rcpp	* 0.12.0	2015-07-25
##	RcppArmadillo	* 0.5.400.2.0	2015-08-17
##	RCurl	1.95-4.7	2015-06-30
##	reshape2	1.4.1	2014-12-06
##	rmarkdown	0.7	2015-06-13
##	rpart	4.1-10	2015-06-29
##	Rsamtools	1.20.4	2015-06-01
##	RSkittleBrewer	* 1.1	2015-09-05
##	RSQLite	* 1.0.0	2014-10-25
##	rstudioapi	0.3.1	2015-04-07
##	rtracklayer	1.28.9	2015-08-19

##	rversions	1.0.2	2015-07-13
##	S4Vectors	* 0.6.5	2015-09-01
##	scales	0.3.0	2015-08-25
##	snm	1.16.0	2015-04-17
##	snpStats	* 1.18.0	2015-04-17
##	stringi	0.5-5	2015-06-29
##	stringr	1.0.0	2015-04-30
##	survival	* 2.38-3	2015-07-02
##	sva	* 3.14.0	2015-04-17
##	tidyr	0.2.0	2014-12-05
##	UsingR	* 2.0-5	2015-08-06
##	whisker	0.3-2	2013-04-28
##	XML	3.98-1.3	2015-06-30
##	xml2	0.1.2	2015-09-01
##	xtable	1.7-4	2014-09-12
##	XVector	0.8.0	2015-04-17
##	yaml	2.1.13	2014-06-12
##	zlibbioc	1.14.0	2015-04-17
##	source		
##	CRAN (R 3.2.0)		
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## CRAN (R 3.2.1)
## Bioconductor
## Github (alyssafranze/RSkittleBrewer@0a96a20)
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It is also useful to compile the time the document was processed. This document was processed on: 2015-09-06.